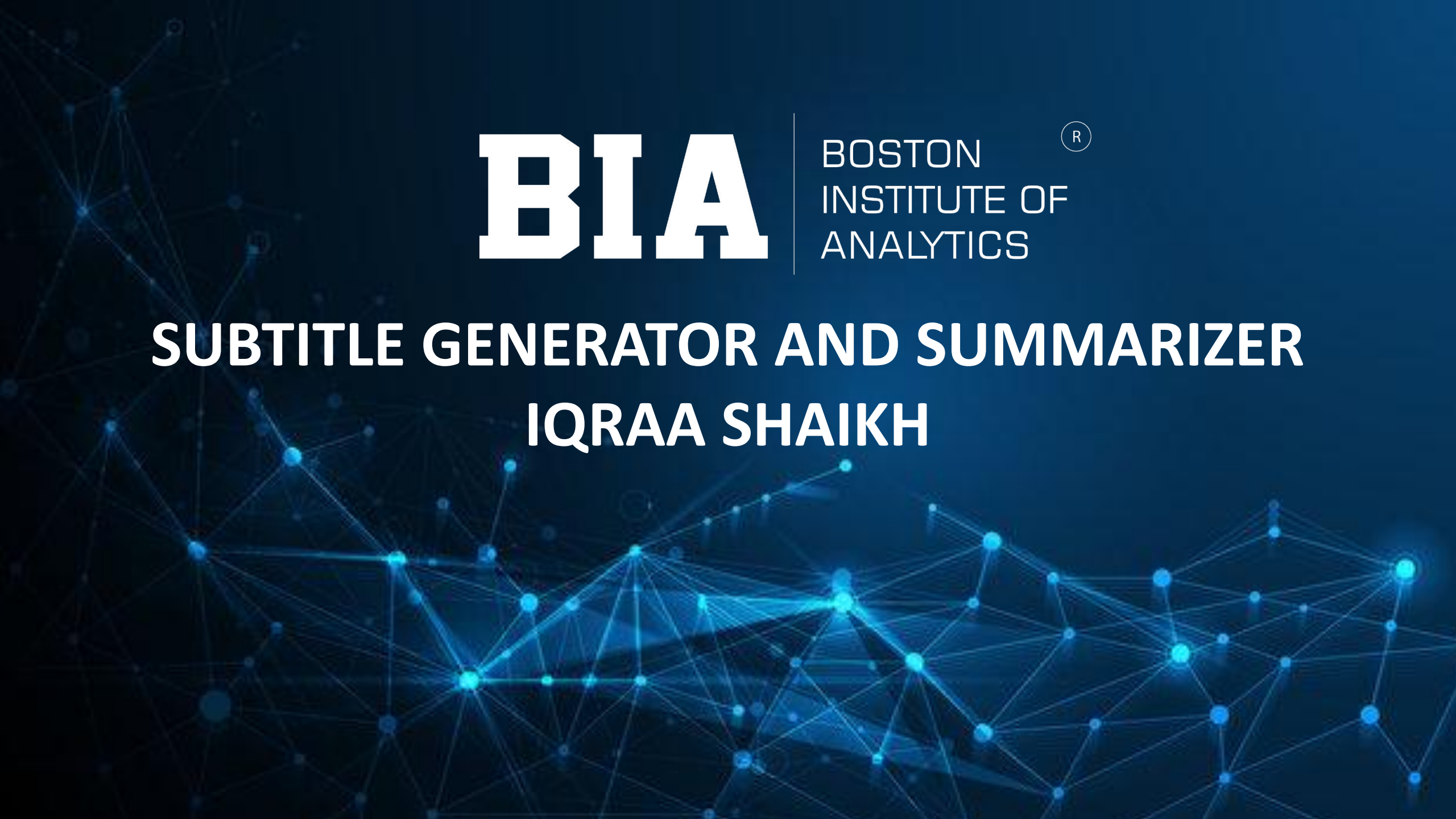


A dark blue background featuring a complex network of glowing blue lines and dots, resembling a molecular or data network structure, with some nodes being brighter than others.

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SUBTITLE GENERATOR AND SUMMARIZER
IQRAA SHAIKH

INTRODUCTION:

- .Online education is growing rapidly, with hundred of lecture videos uploaded monthly.
- .Students often struggle with:
 - .Following long lectures
 - .Accessing key points quickly
 - .Learning in noisy environments or with hearing challenges
- .AI can help automate transcription and summarization to make learning more efficient and accessible.

PROBLEM STATEMENT

- .Online platforms produce numerous lecture videos monthly.
- .Students find it hard to:
 - .Follow long lectures
 - .Access key points quickly
 - .Use videos in noisy environments or for hearing-impaired learning
- .Need: Automated subtitle + summaries for accessibility and engagement.

OBJECTIVES

- .Automated generate subtitles (.srt) from lecture videos.
- .Generator concise summaries (100 words) per video.
- .Make lecture content more accessible and reviewable for students.

SCOPE OF THE PROJECT

- .Extract audio from lecture videos
- .Transcribe audio using Whisper model
- .Summarize transcripts using BART/FLAN-T5 models
- .Generate subtitle files (.srt)
- .Produce short summaries (3-5 sentences)
- .Evaluate using Word Error Rate (WER) and ROUGE scores

PROJECT WORKFLOW OVERVIEW:

1. Video Upload→
2. Audio Extraction→
3. Speech Transcription→
4. Text Summarization→
5. Subtitle Generation→
6. Outputs & Evaluation.

TOOLS & TECHNOLOGIES:

TOOL

Whisper

Transformers(BART)

MoviePy

JiWer

ROUGE

Python/Jupyter

notebook

PURPOSE

Speech-to-text transcription

Text summarization

Audio extraction

WER evaluation

Summary evaluation

Implementation

AUTOMATED SPEECH RECOGNITION (ASR):

Automated Speech Recognition (ASR) is the process of converting spoken language into written text using machine learning models.

Modern ASR system use deep neural networks and transformer architectures to capture complex speech patterns.

One of the most advance ASR systems is whisper, which can transcribe multilingual audio with high accuracy.

KEY COMPONENTS OF ASR:

1. Audio Preprocessing

- .Converts audio signals into spectrograms .
- .Removes noise and normalizes sound.

2. Feature Extraction

- .Extracts important acoustic features such as:
 - .Mel-frequency cepstral coefficients (MFCC)
 - .Spectrogram representations

3. Acoustic Model

- .Maps audio features to phonemes or words.

4. Language Model

- .Predicts word sequences for grammatical correctness.

5. Decoder

- .Produces the final text transcription.

AUDIO EXTRACTION:

- .Extract audio from lecture video using MoviePy
- .Convert video to .wav format for transcription

SPEECH-TO-TEXT TRANSCRIPTION:

- .Use OpenAI Whisper model (base/medium)
- .Generates transcript with timestamps for subtitles

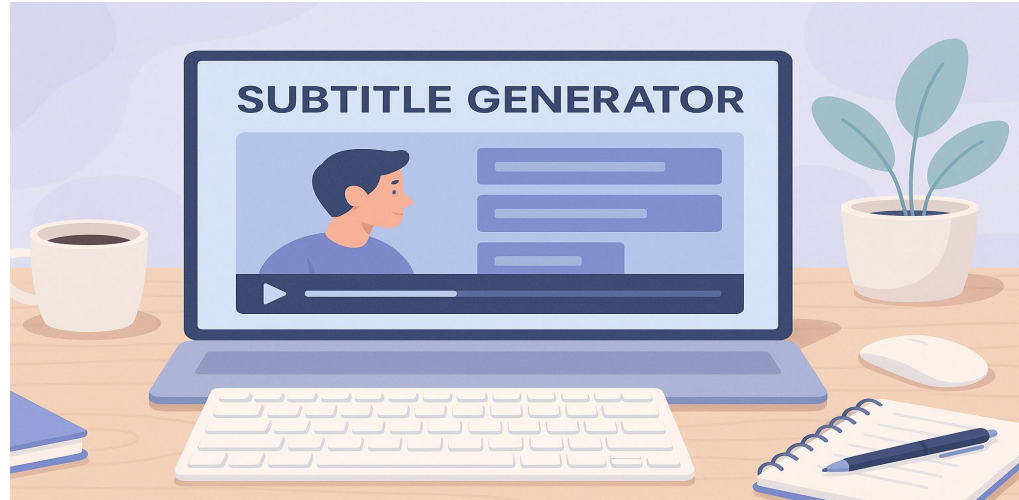
SUBTITLE GENERATION:

Subtitles are textual representation of spoken dialogue synchronized with video timestamp.

The widely used subtitle format is SRT (SubRip Subtitle).

Each subtitle entry contains:

- 1.Subtitle number
- 2.Start timestamp
- 3.End timestamp
- 4.Subtitle text



Using the timestamps generated by whisper, subtitles can be automatically formatted into an SRT file.

TEXT SUMMARIZATION:

Text summarization reduces large volumes of text into shorter versions while preserving key information.

These are the two main approaches:

1. Extractive Summarization

Select important sentences directly from the original text.

Example methods:

- .TextRank

- .TF-IDF sentence ranking

2. Abstractive Summarization

Generates new sentences that capture the main idea of the text

This project uses abstractive summarization.

TRANSFORM MODEL FOR SUMMARIZATION:

Modern summarization system use transformer-based models such as:

- .BART

- .FLAN-T5

BART(Bidirectional and Auto-regressive Transformer)

BART combines two architectures:

- .BERT encoder

- .GPT-like decoder

This allows it to:

- .Understand context

- .Generate fluent summaries

FLAN-T5

Flan-T5 is an instruction-tuned model that performs well across multiple NLP tasks including:

- .Summarization

- .Translation

- .Question answering

OUTPUT:

- .Generated transcripts (.txt)
- .Generated summaries (.txt)
- .Generated subtitles (.srt)
- .All organized per lecture video

EVALUATION:

.Word Error Rate (WER)→ measures transcription accuracy

.Example: $WER=0.11$

.ROUGE Score→ measures summarization quality

.Example: $ROUGE-1 = 0.43$

. $ROUGE-2 = 0.31$

. $ROUGE-L = 0.39$

FUTURE IMPROVEMENTS:

Future enhancements may include:

- .Real-time subtitle generator
- .Multilingual subtitle translation
- .Speaker identification
- .Topic segmentation
- .Key concept extraction

Tools such as WhisperX can improve subtitle alignment and speaker detection.

PROJECT ADVANTAGES:

- .Reduces manual effort in transcribing lecture
- .Improves accessibility for hearing-impaired students
- .Provides concise summaries for faster review
- .Work for any lecture length (scalable solution)

LIMITATION & WORKFLOW:

Limitations:

- .Placeholder videos currently used (short demo videos)
- .Performance depends on audio quality
- .Summarization may miss nuanced details.

Future Work:

- .Real-time transcription & summarization
- .Multilingual support
- .Topic-based summarization for longer lectures

CONCLUSION:

- .Automated subtitles + summaries improves learning efficiency
- .Demonstrated full pipeline from video→ audio→ transcript→ summary→ subtitle
- .Ready for deployment on online education platforms
- .Flexible to handle real lecture videos in future.

The background is a dark blue gradient with a complex network of glowing blue lines and dots, resembling a molecular structure or a data network. The dots are of varying sizes and brightness, with some appearing as bright blue spheres. The lines are thin and connect the dots in a web-like pattern.

Thank You!